OOD Quantification + Alignment → Category-Shift Adapt

Feature / sample / hop-prototype weights with soft$^{\gamma}$ gate

AS-DTA / CFPerm

Abstract

Message. Category shift is an OOD event. We quantify the footprint (RF-domain VIMP + membership), then adapt a CE head via `partial_fit`. v4 adds the project's **alignment → weight** logic (hop / prototype cosine, same family as heatmap IW): rows more aligned with the novel direction than with known prototypes get higher CE mass. Soft$^{\gamma=0.5}$ still tapers empowerment by N_{cap}=200. On text boards, `both+align` leads the scarce window; plain `both` is the more stable endpoint among gated schemes.

1 OOD + alignment → weights

Quantify (OOD). RF on $W \in \{0, 1\}$ (REF vs LIVE) $\Rightarrow$ VIMP v. Top-K subspace S: membership m_i from z-distance to REF.

Align (prototype / hop cosine). On S, let μ_c be known-class prototypes and μ_{nov} the cold-start novel mean. For each live row

$$\text{align}_i = \sigma\big(\beta\,(\cos(x_i, \mu_{\text{nov}}) - \max_c \cos(x_i, \mu_c))\big). \tag{1}$$

High align$_i$ = aligned with the new-category direction more than with old centers — the same "high alignment → upweight" rule as hop-cosine IW on the batch heatmap.

Adapt (three cheap channels).

- **Feature:** $x_{ij} \leftarrow x_{ij}(1 + \alpha_{\text{eff}} \tilde{v}_j)$ (optional blend with $|\mu_{\text{nov}} - \mu_{\text{ref}}|$).
- **Sample:** membership fused excess.
- **Alignment:** excess $+ \lambda_a \,\text{align}_i$.

All excesses are multiplied by the soft$^{\gamma}$ cold-start decay $((1 - n_{\text{idx}}/200)_+)^{0.5}$, then passed to `partial_fit(..., sample_weight=`$w$`)`. No new backbone, no EWMA.

2 Curves (FULL LIVE holdout)

Numbers.

	uniform	both	align	both+align
acc @ b=20/cat	0.456	0.475	0.468	**0.478**
endpoint acc	0.636	**0.647**	0.641	0.641
endpoint Δacc	—	**+0.011**	+0.005	+0.005

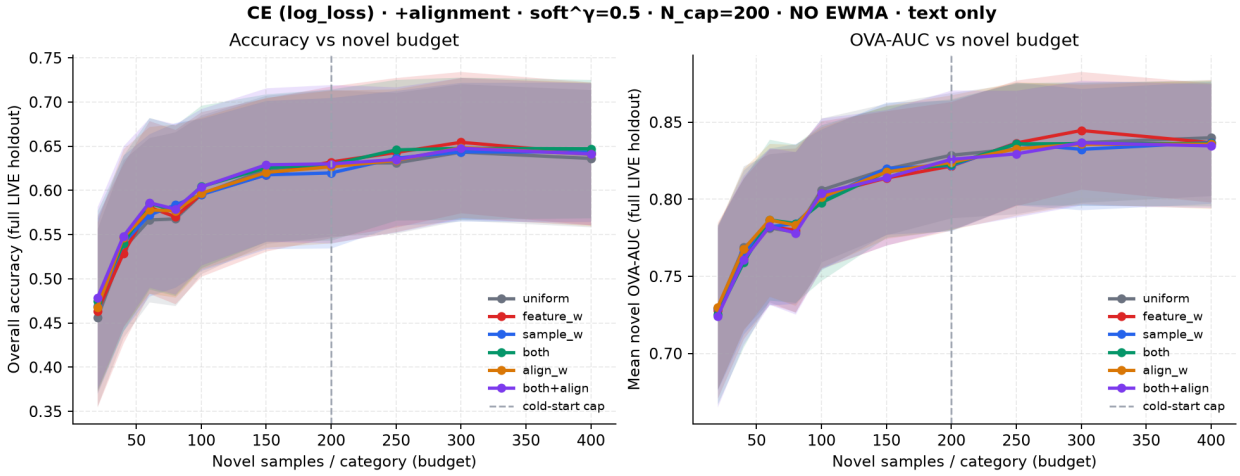


Figure 1: Soft$^{\gamma=0.5}$ schemes including alignment (mean±SEM). Scarce regime: `both+align` leads. Past the cap: gated schemes stay near / above uniform.

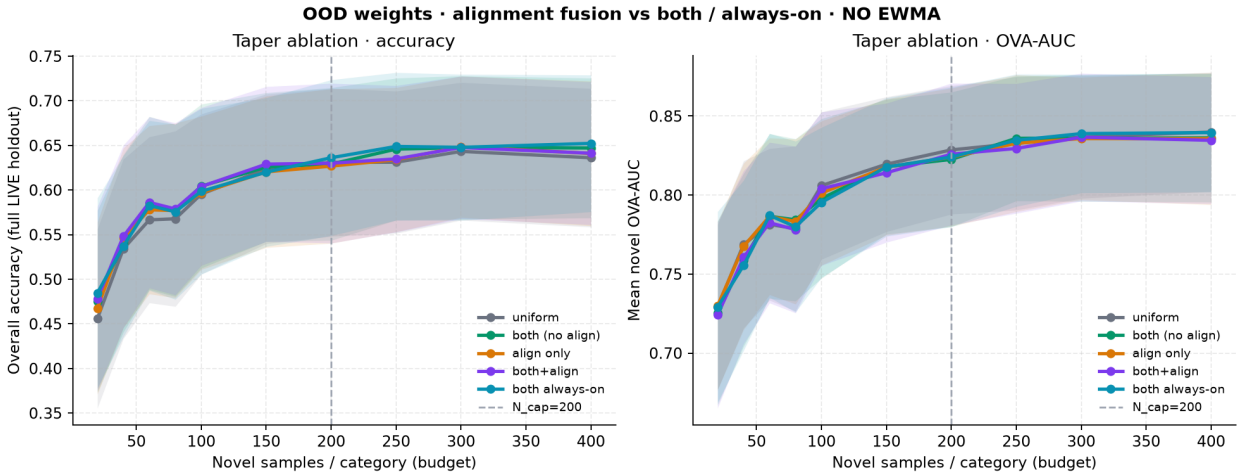


Figure 2: Alignment ablation: `both`, `align` only, `both+align`, always-on. Alignment helps when the new category is scarce; plain `both` is the stabler gated endpoint.

3 Ops rule

1. **Very scarce** new category: prefer `both+align` (early lift).
2. **Default gated online:** `both` with soft$^{\gamma=0.5}$ (best gated endpoint).
3. Alignment is the same hop-cosine family already used elsewhere — plug into `sample_weight`, keep the head.